

Risk Categories

4 categories :

- Violent
- Property
- Public Safety / Accident Risk
- Women safety

! Violent :

Murder , Attempt to Murder , Rape , Assault , Riot

! Property :

– Robbery , Theft , House Theft , Burglary , Snatching , Motor vehicle theft.

! Public safety / Accident Risk :

– Fatal accident , simple accident , injure count , killed count

! Women safety :

– Rape , Assault (women) , kidnap (women) , cruelty by husband
,Dowry death

ML-models :

- Crime Prediction model
- Risk Classification model
- Resource allocation optimization

Important Insights :

① Crime Trend Insight

- Total IPC crimes increased 8% (2021 → 2022)
Highlight of State of Policing ...
- House Breaking ↑ 315% (2021 to 2022 till July)
Highlight of State of Policing ...

- Attempt to Murder ↑ 31%
 - Molestation ↑ 19%
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2 Women Safety — Strong Research Angle

- 49% rape cases 2022 me minor girls ke against
Highlight of State of Policing ...
- POCSO rape cases 2018 → 2022: 643 → 829 (29% increase)
Highlight of State of Policing ...

women_safety_risk ko split kar sakte ho:

- adult_cases
 - minor_cases.
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3 District-Level Variation (Super Important)

- West district major crimes ↑ 1186% (2021→2022)
Highlight of State of Policing ...
 - Shahdara ↑ 739%
 - East ↑ 531%
 - South East 2022 me multiple crime heads me highest
Highlight of State of Policing ...
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4 Forensic Bottleneck — MASSIVE RESEARCH OPPORTUNITY

- 67% forensic staff shortage
Highlight of State of Policing ...
- Only 66% cases tested
Highlight of State of Policing ...

Crime resolution delay → deterrence weak → repeat crime risk ↑

5 Police Wellbeing Data Missing

50% districts ailment data maintain hi nahi karte

Highlight of State of Policing ...

This shows:

- Data governance gap
- Monitoring weakness

Database :

Columns that we have :

1 Final Master Dataset (Synthetic + Institutional)

Isme columns:

- district
- year
- population
- violent_cases
- property_cases
- women_safety_cases
- accident_cases
- unemployment_rate
- traffic_density_index
- economic_activity_index
- police_units_per_100k

- forensic_capacity_ratio
 - forensic_staff_shortage_index
 - case_processing_delay_index
 - data_transparency_flag
 - shock_year_flag
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Accident Real Dataset

- YEAR
 - DISTRICT
 - TYPE OF ACCIDENT
 - **INJURED**
 - **KILLED**
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Step 2: Logical Grouping :

A. Identity / Static Info

- district
- year

Ye primary key level info hai.

B. Demographic

- population
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C. Crime Related

- violent_cases
- property_cases
- women_safety_cases
- accident_cases

(Real accident merge hone ke baad refined ho sakta hai)



D. Socio-Economic Drivers

- unemployment_rate
 - traffic_density_index
 - economic_activity_index
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E. Institutional Capacity

- police_units_per_100k
 - forensic_capacity_ratio
 - forensic_staff_shortage_index
 - case_processing_delay_index
 - data_transparency_flag
-



F. External Shock

- shock_year_flag

